Polar Equations and Graphs

ANSWER KEY

Recognizing polar graphs

1 Identify the graph of each polar equation:

a $r = 3$ Circle of radius 3 centered at the pole.

b $\theta = \frac{\pi}{6}$ Line through the pole at angle $\frac{\pi}{6}$.

c $r = 2 + 2\cos\theta$ Cardioid (heart-shaped, symmetric about the polar axis).

d $r = 4\sin 3\theta$ Rose curve with 3 petals.

2 For the limaçon $r = 1 + 2\cos\theta$:

a Explain how you know it has an inner loop. In $r = a + b\cos\theta$, $|a| < |b|$ (here $1 < 2$) produces an inner loop.

b Find the values of θ in $[0, 2\pi)$ at which the curve passes through the pole. $1 + 2\cos\theta = 0$ gives $\cos\theta = -\frac{1}{2}$: $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$.

3 For the rose $r = 5\sin 2\theta$:

a State the number of petals. $n = 2$ is even, so $2n = 4$ petals.

b State the maximum value of $|r|$ and the smallest positive θ at which it occurs. Maximum $|r| = 5$, first at $2\theta = \frac{\pi}{2}$, i.e. $\theta = \frac{\pi}{4}$.

4 Determine whether the graph of $r = 2 + \sin\theta$ passes through the pole, and state the type of limaçon. $2 + \sin\theta \geq 1 > 0$ for all θ, so it never passes through the pole. Since $|a| > |b|$ $(2 > 1)$, it is a dimpled limaçon with no inner loop.